

Lec 10-1, Nov 10, 2025.

- Inverse trigonometric functions (7.6).

- Integration by parts (8.2)

 - ↳ Indefinite integrals

 - ↳ Reduction formulae

 - ↳ Definite integrals

 - ↳ Tabular integration

- Trigonometric integrals (8.3)

 - $\int \sec^n x \, dx$

 - $\int \tan^m x \sec^n x \, dx$

 - Integrands with $\sqrt{1 \pm \cos 2\theta}$

 - $\int \sin mx \sin nx \, dx$, $\int \sin mx \cos nx \, dx$, $\int \cos mx \cos nx \, dx$.

Inverse trigonometric functions

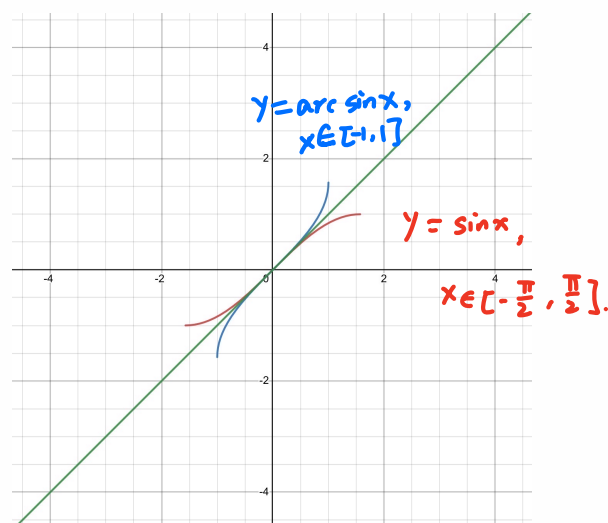
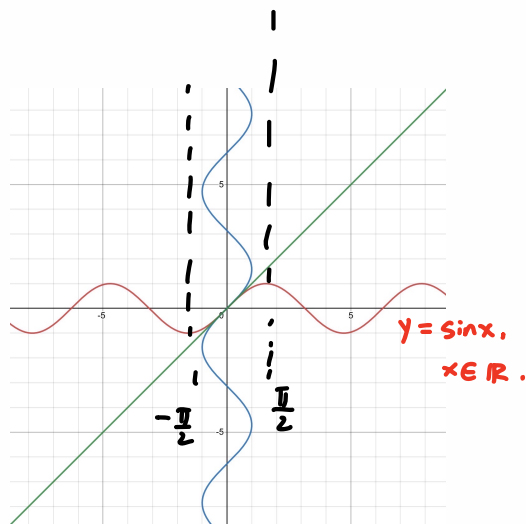
(one-to-one)

The $\sin$ function is not injective on its natural domain $\mathbb{R}$, so it does not have an inverse in this case.

But $\sin: [-\frac{\pi}{2}, \frac{\pi}{2}] \xrightarrow{\sin} [-1, 1]$ is increasing, so it is injective and has an inverse $\sin^{-1}$.

$$\text{Arc sin} : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}].$$

This is the inverse sine function, which is also denoted by $\sin^{-1}$.



Q: $\arcsin'(x) = ?$. Let $f(x) = \sin x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. $\Rightarrow$

Sol': $(f^{-1})'(y) \stackrel{\text{Thm}}{=} \frac{1}{f'(x)} \Big|_{x=f^{-1}(y)} \begin{cases} \sin x = y \\ x = \arcsin y \end{cases} \xrightarrow{\downarrow} \cos x > 0$

$$= \frac{1}{\cos(x)} \Big|_{x=f^{-1}(y)} \Rightarrow \cos(\arcsin y) = \cos x$$

$$= \frac{1}{\cos(\arcsin(y))} \Big| = \sqrt{1 - \sin^2 x}$$

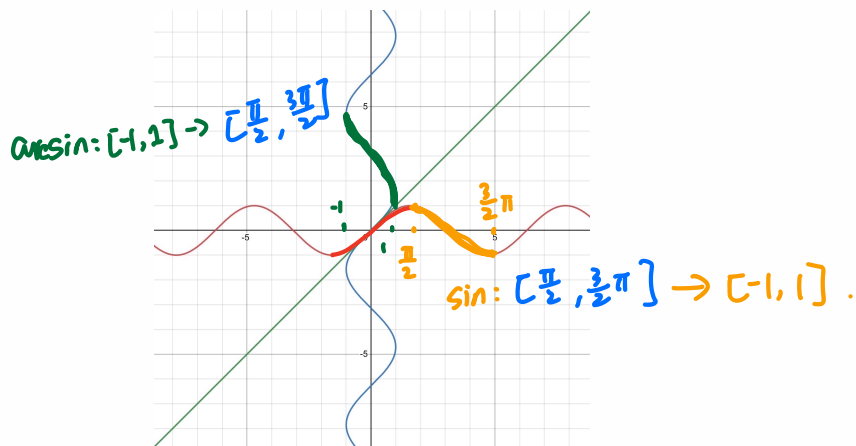
$$= \frac{1}{\sqrt{1 - y^2}}, \forall y \in (-1, 1).$$

Hence, $\arcsin'(y) = \frac{1}{\sqrt{1 - y^2}}, \forall y \in (-1, 1).$ (*) $\square$

Rmks. • $\arcsin'(x)$ does not exist at $x = \pm 1$ (infinite slope).

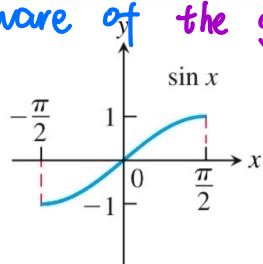
• The formula (*) assumes that the domain of sine is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

It may be different otherwise.

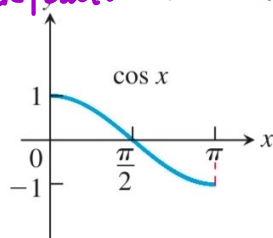


The following figures summarize the six trigonometric functions and their inverses.

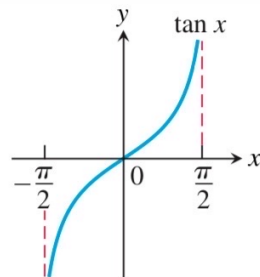
Be aware of the given default domains.



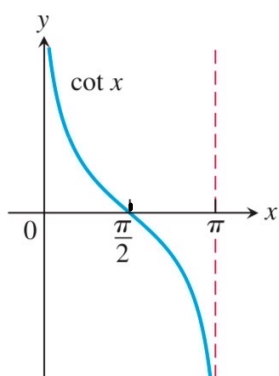
$y = \sin x$
 Domain: $[-\pi/2, \pi/2]$
 Range: $[-1, 1]$



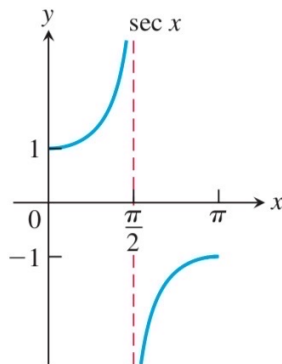
$y = \cos x$
 Domain: $[0, \pi]$
 Range: $[-1, 1]$



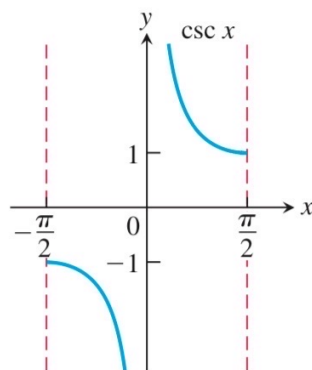
$y = \tan x$
 Domain: $(-\pi/2, \pi/2)$
 Range: $(-\infty, \infty)$



$y = \cot x = \frac{1}{\tan x}$
 Domain: $(0, \pi)$
 Range: $(-\infty, \infty)$

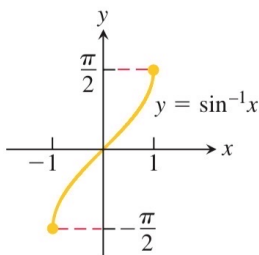


$y = \sec x = \frac{1}{\cos x}$
 Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
 Range: $(-\infty, -1] \cup [1, \infty)$



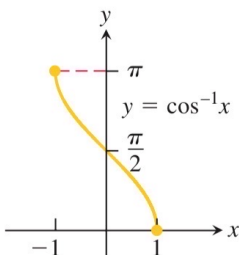
$y = \csc x = \frac{1}{\sin x}$
 Domain: $(-\pi/2, 0) \cup (0, \pi/2]$
 Range: $(-\infty, -1] \cup [1, \infty)$

Domain: $-1 \leq x \leq 1$
 Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



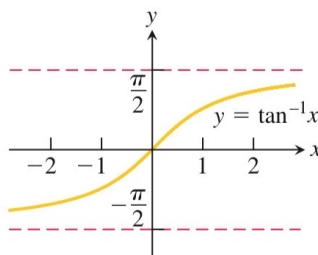
(a)

Domain: $-1 \leq x \leq 1$
 Range: $0 \leq y \leq \pi$



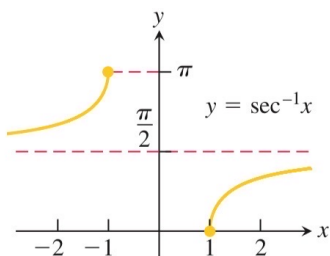
(b)

Domain: $-\infty < x < \infty$
 Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



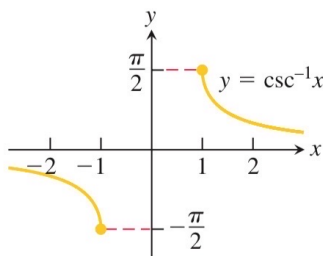
(c)

Domain: $x \leq -1$ or $x \geq 1$
 Range: $0 \leq y \leq \pi, y \neq \frac{\pi}{2}$



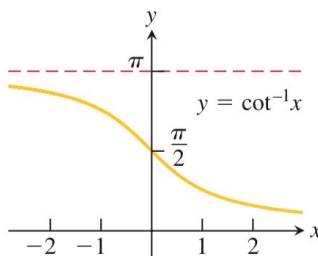
(d)

Domain: $x \leq -1$ or $x \geq 1$
 Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$



(e)

Domain: $-\infty < x < \infty$
 Range: $0 < y < \pi$



(f)

We can use inverse differentiation to obtain the derivative formulae for other trigonometric functions.

e.g. For $f: \tan: (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$, the inverse function is

$$\arctan: \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2}) .$$

For $y_0 \in \mathbb{R}$ and $\tan x_0 = y_0$, we have

$$\begin{aligned} \arctan'(y_0) &\stackrel{\text{Thm}}{=} \frac{1}{f'(x_0)} \Big|_{x_0 = f^{-1}(y_0)} \\ (f^{-1})'(y_0) &= \frac{1}{\frac{1}{\cos^2 x_0}} \Big|_{x_0 = f^{-1}(y_0)} \\ &= \cos^2 x_0 \\ &= \frac{1}{1+y_0^2}, \quad y_0 \in \mathbb{R} \end{aligned}$$

Hence,

$$\boxed{\arctan'(y) = \frac{1}{1+y^2}, \quad y \in \mathbb{R} .}$$

$$\begin{aligned} &\left\{ \begin{array}{l} \tan x_0 = y_0 \\ x_0 = \arctan y_0 \end{array} \right. \\ &\Rightarrow \cos^2(\arctan y_0) \\ &= \cos^2(x_0) = \frac{1}{1+y_0^2} \\ &\left\{ \begin{array}{l} \cos^2(x_0) + \sin^2(x_0) = 1 \\ \frac{\sin(x_0)}{\cos(x_0)} = y_0 \end{array} \right. \\ &\Rightarrow \cos^2(x_0)(1 + \tan^2 x_0) = 1 \\ &\Rightarrow \cos^2(x_0)(1 + y_0^2) = 1 \\ &\Rightarrow \cos^2(x_0) = \frac{1}{1+y_0^2} \quad \square \end{aligned}$$

e.g. For $\sec: [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi] \rightarrow (-\infty, -1] \cup [1, \infty)$, we have

the inverse: $\operatorname{arcsec}: (-\infty, -1] \cup [1, \infty) \rightarrow [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$.

Let $f(x) = \sec(x) = \frac{1}{\cos(x)}$, $x \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$.

$$\begin{aligned}
 \therefore (f^{-1})'(y) &\stackrel{\text{thm}}{=} \frac{1}{f'(x)} \Big|_{x=f^{-1}(y)} \\
 &= \frac{1}{\left(\frac{1}{\cos x}\right)'} \Big|_{x=f^{-1}(y)} \\
 &= \frac{1}{\frac{\sin x}{\cos^2 x}} \Big|_{x=f^{-1}(y)} \\
 &= \frac{\cos^2 x}{\sin x} \Big|_{x=f^{-1}(y)} \\
 &= \frac{1}{|y| \sqrt{y^2-1}}, \quad \forall y \in (-\infty, -1) \cup (1, \infty).
 \end{aligned}$$

$$\begin{aligned}
 y_0 &\in (-\infty, -1] \cup [1, \infty) \\
 \Rightarrow x_0 &\in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi] \\
 \Rightarrow \sin x_0 &\geq 0
 \end{aligned}$$

$$\begin{cases} \sec(x_0) = \frac{1}{\cos x_0} = y_0 \\ x_0 = \arccos \frac{1}{y_0} \end{cases}$$

$$\begin{aligned}
 \Rightarrow \cos x_0 &= \frac{1}{y_0} \\
 \Rightarrow \sin x_0 &= \sqrt{1 - \left(\frac{1}{y_0}\right)^2} \\
 \Rightarrow \frac{\cos^2 x_0}{\sin x_0} &= \frac{\left(\frac{1}{y_0}\right)^2}{\sqrt{1 - \left(\frac{1}{y_0}\right)^2}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{|y_0| |y_0| \sqrt{1 - \frac{1}{y_0^2}}} \\
 &= \frac{1}{|y_0| \sqrt{y_0^2 - \frac{1}{y_0^2} \cdot y_0^2}} \\
 &= \frac{1}{|y_0| \sqrt{y_0^2 - 1}}
 \end{aligned}$$

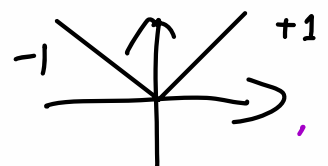
Hence, $\arccos'(y) = \frac{1}{|y| \sqrt{y^2-1}}, \quad \forall y \text{ with } |y| > 1.$

□.

Further, as $g(x) = |x|$ has a derivative $g'(x) = \frac{|x|}{x}, x \neq 0$, then let $f(x) = \arccos(x)$,

$$\frac{d}{dx} [\arccos(|x|)] = \frac{d}{dx} [f(g(x))] = f'(g(x))g'(x) = \frac{1}{|g(x)| \sqrt{g(x)^2-1}} \cdot \frac{|x|}{x} = \frac{1}{|x| \sqrt{x^2-1}} \cdot \frac{|x|}{x} = \frac{1}{x \sqrt{x^2-1}}$$

i.e. $\frac{d}{dx} [\arccos(|x|)] = \frac{1}{x \sqrt{x^2-1}}, \quad \forall x \text{ with } |x| > 1.$



Others

$$\cos(?) = x_0 = \sin y_0 = \cos\left(\frac{\pi}{2} - y_0\right)$$



$$? = \frac{\pi}{2} - y_0$$

Since $\sin(y) = \cos\left(\frac{\pi}{2} - y\right)$, if $x_0 = \sin y_0$, then

$$\arcsin x_0 = y_0$$

$$\text{and } \arccos x_0 = ? = \frac{\pi}{2} - y_0.$$

So

$$\boxed{\overbrace{\arcsin x}^{y_0} + \overbrace{\arccos x}^{\frac{\pi}{2} - y_0} = \frac{\pi}{2}, \quad \forall x \in [-1, 1].}$$

Similarly, $\arctan x_0 = y_0$ and $\operatorname{arccot} x_0 = \frac{\pi}{2} - y_0$.

$$\left[\because \tan y_0 = x_0 \right]$$

$$\cot\left(\frac{\pi}{2} - y_0\right) = \frac{1}{\tan\left(\frac{\pi}{2} - y_0\right)} = \frac{1}{\frac{1}{\tan(y_0)}} = \tan(y_0) = x_0 \Rightarrow \frac{\pi}{2} - y_0 = \operatorname{arccot} x_0.$$

So

$$\left(\tan \frac{\pi}{6} = \frac{1}{\tan \frac{\pi}{3}} = \frac{\sqrt{3}}{3}\right)$$

$$\boxed{\underbrace{\arctan x}_{y_0} + \underbrace{\operatorname{arccot} x}_{\frac{\pi}{2} - y_0} = \frac{\pi}{2}, \quad \forall x \in (-\infty, 0) \cup (0, \infty).}$$

Similarly,

$$\boxed{\underbrace{\operatorname{arcsec} x}_{\Delta} + \underbrace{\operatorname{arccsc} x}_{\Delta} = \frac{\pi}{2}, \quad \forall x \in (-\infty, -1] \cup [1, \infty).}$$

Summary:

$$\triangleright \sin : (-\pi/2, \pi/2) \rightarrow (-1, 1), \arcsin'(x) = \frac{1}{\sqrt{1-x^2}}.$$

$$\triangleright \cos : (0, \pi) \rightarrow (-1, 1), \arccos'(x) = \frac{-1}{\sqrt{1-x^2}}.$$

$$\triangleright \tan : (-\pi/2, \pi/2) \rightarrow \mathbb{R}, \arctan'(x) = \frac{1}{1+x^2}.$$

$$\triangleright \cot : (0, \pi) \rightarrow \mathbb{R}, \operatorname{arccot}'(x) = \frac{-1}{1+x^2}.$$

$$\triangleright \sec : (0, \pi/2) \cup (\pi/2, \pi) \rightarrow (-\infty, -1) \cup (1, \infty),$$

$$\operatorname{arcsec}'(x) = \frac{1}{|x|\sqrt{x^2-1}}.$$

$$\triangleright \csc : (-\pi/2, 0) \cup (0, \pi/2) \rightarrow (-\infty, -1) \cup (1, \infty),$$

$$\operatorname{arccsc}'(x) = \frac{-1}{|x|\sqrt{x^2-1}}.$$

$$\frac{1}{\sqrt{-ax^2+bx+c}} \quad (2)$$

$$\frac{1}{ax^2+bx+c} \quad (1)$$

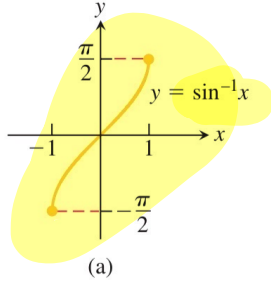
$$\frac{1}{|x|\sqrt{ax^2+bx+c}} \quad (4)$$

$$\frac{1}{x\sqrt{ax^2+bx+c}} \quad (3)$$

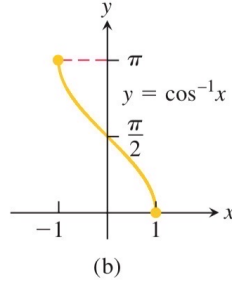
We further have

$$0 = \frac{d}{dx} \left(\frac{\pi}{2} \right) = \arcsin' x + \arccos' x = \arctan' x + \operatorname{arccot}' x = \operatorname{arcsec}' x + \operatorname{arccsc}' x.$$

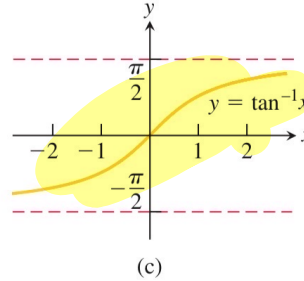
Domain: $-1 \leq x \leq 1$
Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



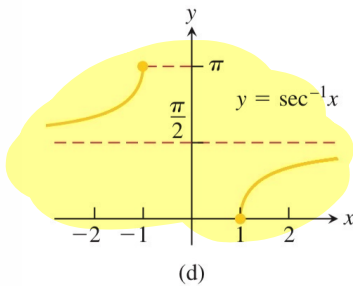
Domain: $-1 \leq x \leq 1$
Range: $0 \leq y \leq \pi$



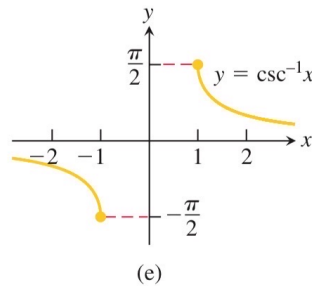
Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



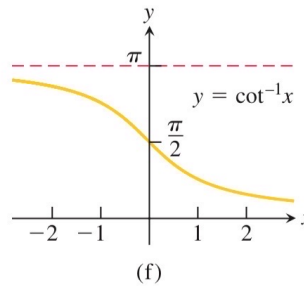
Domain: $x \leq -1$ or $x \geq 1$
Range: $0 \leq y \leq \pi, y \neq \frac{\pi}{2}$



Domain: $x \leq -1$ or $x \geq 1$
Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$



Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$



e.g. $1. \int \frac{dx}{\sqrt{3-4x^2}} = ?$

Sol: LHS = $\frac{1}{\sqrt{3}} \int \frac{dx}{\sqrt{1-\frac{4}{3}x^2}}$

$u = \frac{2}{\sqrt{3}}x$
 $x = \frac{\sqrt{3}}{2}u$

$$\frac{1}{\sqrt{3}} \int \frac{1}{\sqrt{1-u^2}} du \Big|_{u=\frac{2}{\sqrt{3}}x} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{1}{\sqrt{3}} \frac{\sqrt{3}}{2} \int \frac{1}{\sqrt{1-u^2}} du \Big|_{u=\frac{2}{\sqrt{3}}x}$$

$$= \frac{1}{2} \operatorname{arcsin}(u) \Big|_{u=\frac{2}{\sqrt{3}}x} + C = \frac{1}{2} \operatorname{arcsin}\left(\frac{2}{\sqrt{3}}x\right) + C. \quad \square$$

$$2. \int \frac{dx}{4x^2+4x+2} = ?$$

$$\text{Sol: LHS} = \int \frac{dx}{4(x+\frac{1}{2})^2+1}$$

$$\begin{aligned} u &= 2(x+\frac{1}{2}) \\ &= \\ x &= \frac{u}{2} - \frac{1}{2} \end{aligned} \int \frac{1}{u^2+1} d\left(\frac{u-1}{2}\right) \Big|_{u=2x+1}$$

$$= \frac{1}{2} \int \frac{1}{u^2+1} du \Big|_{u=2x+1}$$

$$= \frac{1}{2} \arctan u \Big|_{u=2x+1} + C$$

$$= \frac{1}{2} \arctan(2x+1) + C.$$

□.

$$3. \int \frac{dx}{x\sqrt{x^2-a^2}} = ?$$

$$\text{Sol: LHS} = \int \frac{dx}{x|a|\sqrt{(\frac{x}{a})^2-1}}$$

$$\begin{aligned} \frac{x}{a} &= u \\ &= \\ x &= au \end{aligned} \int \frac{1}{a|a|\sqrt{u^2-1}} d(au) \Big|_{u=\frac{x}{a}}$$

$$\begin{aligned} &= \frac{a}{a|a|} \arccos(|u|) \Big|_{u=\frac{x}{a}} + C = \frac{1}{|a|} \arccos(|u|) \Big|_{u=\frac{x}{a}} + C \\ &= \frac{1}{|a|} \arccos\left(\left|\frac{x}{a}\right|\right) + C. \quad \square. \end{aligned}$$

- Integration by Parts : A technique to simplify integrals.

— Indefinite integrals

If f and g are differentiable at x , then

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x),$$

so

$$\int (f'(x)g(x) + f(x)g'(x)) dx = f(x)g(x) + C.$$

so

Formula (Integration by Parts)

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

The "+C" above is absorbed by this.

Example

- (i) To find $\int x \cos x dx$, consider $f(x) := x$ and $g'(x) := \cos x$.

We would like to integrate $f(x)g'(x)$. Note that $f'(x) = 1$, and $g(x) = \sin x$ is one antiderivative of g' . Now $f'(x)g(x) = \sin x$ is easy to integrate, and

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C.$$

$$\begin{aligned} \int x \cos x dx &= \int x (g(x))' dx = x \sin x - \int x' \sin x dx \\ (= \int x d \sin x) &= x \sin x - \int \sin x dx \\ &= x \sin x + \cos x + C \end{aligned}$$

□.

If we consider $u=f(x)$ and $v=g(x)$, then the formula above can be written in an alternative form:

$$\int u \, dv = u \cdot v - \int v \, du$$

Example:

(ii) $\int x^2 e^x \, dx = ?$

(iii) $\int e^x \sin x \, dx = ?$

(iv) $\int \ln x \, dx = ?$

Sol: (ii) LHS = $\int x^2 (e^x)' \, dx = x^2 e^x - \int (x^2)' e^x \, dx$
 $= x^2 e^x - 2 \int x e^x \, dx = x^2 e^x - 2 [\int x (e^x)' \, dx]$
 $= x^2 e^x - 2 [x e^x - \int x' e^x \, dx] = x^2 e^x - 2 [x e^x - \int e^x \, dx]$
 $= x^2 e^x - 2 [x e^x - e^x] + C = x^2 e^x - 2x e^x + 2e^x + C$, where C is any real number. 5

(iii) LHS = $\int (e^x)' \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx$
 $= e^x \sin x - [\int (e^x)' \cos x \, dx]$
 $= e^x \sin x - [e^x \cos x - \int e^x (\cos x)' \, dx]$
 $= e^x \sin x - [e^x \cos x + \int e^x \sin x \, dx]$
 $= e^x \sin x - e^x \cos x - \int e^x \sin x \, dx$
 $= e^x \sin x - e^x \cos x - \text{LHS} + C$

$\Rightarrow 2\text{LHS} = e^x \sin x - e^x \cos x + C$

$$\Rightarrow \text{LHS} = \frac{1}{2}(e^x \sin x - e^x \cos x) + C$$

□

$$\begin{aligned} \text{[check: } \left[\frac{1}{2} e^x (\sin x - \cos x) \right]' &= \frac{1}{2} e^x (\sin x - \cos x) + \frac{1}{2} e^x (\cos x + \sin x) \\ &= \frac{1}{2} e^x \cdot 2 \sin x = e^x \sin x. \quad] \quad \checkmark \end{aligned}$$

$$\text{iv) LHS} = \int 1 \cdot \ln x \, dx = \int x' \cdot \ln x \, dx$$

$$= x \ln x - \int x (\ln x)' \, dx$$

$$= x \ln x - \int x \frac{1}{x} \, dx$$

$$= x \ln x - x + C$$

□

Reduction Formulae

Consider finding $I = \int \sin^n x \, dx$, where $n = 1, 2, 3, \dots$

Step 1. Set $u = \sin^{n-1} x$. Set $v = -\cos x$ and $dv = \sin x \, dx$.

$$\text{Then } I = \int \sin^{n-1} x (\sin x) \, dx = \int \sin^{n-1} x \, d(-\cos x) = \int u \, dv.$$

$$\text{Step 3. } I = \int \sin^n x \, dx = \int u \, dv$$

$$= u v - \int v \, du = \sin^{n-1}(x) (-\cos x) - \int (-\cos x) d(\sin^{n-1} x)$$

$$= \sin^{n-1}(x) (-\cos x) - \int (-\cos x) (n-1) \sin^{n-2}(x) \cdot \cos x \, dx$$

$$= -\sin^{n-1}(x) \cos x + (n-1) \int \cos^2 x \sin^{n-2}(x) \, dx$$

$$= -\sin^{n-1}(x) \cos x + (n-1) \int (1 - \sin^2 x) \sin^{n-2}(x) dx$$

$$= -\sin^{n-1}(x) \cos x + (n-1) \int \sin^{n-2}(x) dx - (n-1) \underbrace{\int \sin^n x dx}_I$$

$$\Rightarrow I = -\sin^{n-1}(x) \cos x + (n-1) \int \sin^{n-2}(x) dx - (n-1) \cdot I$$

Step 3 . $n I = -\sin^{n-1}(x) \cos x + (n-1) \int \sin^{n-2}(x) dx$

$$\Rightarrow \underbrace{I}_{\int \sin^n x dx} = -\frac{1}{n} \sin^{n-1}(x) \cos x + \frac{n-1}{n} \int \sin^{n-2}(x) dx$$

This gives the following reduction formula.

$$\boxed{\int \sin^n x dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x dx. \quad \checkmark \quad (*)}$$

Similarly, we have

$$\boxed{\int \cos^n x dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} \int \cos^{n-2} x dx. \quad (**)}$$

e.g. (v) $\int \sin^3 x = ?$

Sol': LHS = $-\frac{1}{3} \cos x \sin^2 x + \frac{3-1}{3} \int \sin x dx$

$$= -\frac{1}{3} \cos x \sin^2 x + \frac{2}{3} (-\cos x) + C \quad \square$$

Rmk: No need to memorize (*) and (**). Just derive by yourself using integration by parts when facing such a question.

$$\begin{aligned}\int \sin^3 x \, dx &= \int \sin^2 x \cdot \sin x \, dx \\&= - \int \sin^2 x (\cos x)' \, dx \\&= - \left[\sin^2 x \cdot \cos x - \int (\sin^2 x)' \cos x \, dx \right] \\&= - \left[\sin^2 x \cos x - \int 2 \sin x \cos x \cos x \, dx \right] \\&= - \sin^2 x \cos x + \int 2 \cos^2 x \sin x \, dx \\&= - \sin^2 x \cos x + \int 2(1 - \sin^2 x) \sin x \, dx \\&= - \sin x \cos x + \int 2 \sin x \, dx - 2 \int \sin^3 x \, dx .\end{aligned}$$

$$\Rightarrow \int \sin^3 x \, dx = \frac{1}{3} [-\sin x \cos x + \int 2 \sin x \, dx] .$$

- Definite integrals

If $f(x)g'(x) + f'(x)g(x)$ is continuous on $[a, b]$, then by FTC 2,

$$\int_a^b (f(x)g'(x) + f'(x)g(x)) \, dx = [f(x)g(x)] \Big|_a^b .$$

Hence,

$$\boxed{\int_a^b f(x)g'(x) \, dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx .}$$

e.g. (vi) $\int_0^1 \arctan x \, dx = ?$

Sol¹: LHS = $\int_0^1 x' \cdot \arctan x \, dx$

$$= x \arctan x \Big|_0^1 - \int_0^1 x (\arctan x)' \, dx$$

$$= 1 \arctan 1 - 0 - \int_0^1 \frac{x}{1+x^2} \, dx$$

$$= 1 \arctan 1 - 0 - \frac{1}{2} \int_0^1 \frac{1}{1+x^2} d(x^2)$$

$u = x^2$

$$= 1 \arctan 1 - 0 - \frac{1}{2} \int_0^1 \frac{1}{1+u} \, du$$

$$= 1 \arctan 1 - 0 - \frac{1}{2} \ln |1+u| \Big|_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} (\ln 2 - \ln 1)$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2.$$

□

- Tabular integration

Consider finding $\int x^{10} \cos x \, dx = \int f_0(x) g_0(x) \, dx$

- You may apply integration by parts 10 times, although you probably would rather spend your time on something else.

• Tabular integration can simplify this process.

$$\begin{aligned} \int \underbrace{f_0(x)}_{g'_1(x)} g_0(x) \, dx &= f_0(x) g_1(x) - \int f'_0(x) g_1(x) \, dx \quad [g'_1 = g_0] \\ &= f_0(x) g_1(x) - \int f_1(x) g_1(x) \, dx \quad [f'_0 = f_1] \\ &= f_0(x) g_1(x) - [f_1(x) g_2(x) - \int f'_1(x) g_2(x) \, dx] \quad [g'_2 = g_1] \\ &= f_0(x) g_1(x) - [f_1(x) g_2(x) - \int f_2(x) g_2(x) \, dx] \quad [f'_1 = f_2] \\ &= f_0(x) g_1(x) - f_1(x) g_2(x) + \int f_2(x) g_2(x) \, dx \\ &= \dots \\ &= f_0(x) g_1(x) - f_1(x) g_2(x) + f_2(x) g_3(x) - \dots \\ &\quad \pm \int f_m(x) g_m(x) \, dx \end{aligned}$$

where ... differentiation: $f'_i(x) = f_{i+1}(x) \quad [f_0(x) = x^{10}]$
 $f_0(x) \rightarrow f_1(x) \rightarrow f_2(x) \rightarrow \dots \rightarrow f_m(x) = 0$

$g_0(x) \rightarrow g_1(x) \rightarrow g_2(x) \rightarrow \dots \rightarrow g_m(x)$

... integration: $\int g_i(x) \, dx = g_{i+1}(x) \quad [g_0(x) = \cos x]$

This method is particularly effective for $\int x^n g(x) dx$.

e.g. (viii) $\int \underbrace{x^3}_{f_0(x)} \underbrace{3^x}_{g_0(x)} dx = ?$

$$= x^3 \frac{3^x}{\ln 3} - \int 3x^2 \frac{3^x}{\ln 3} dx$$

$$= x^3 \frac{3^x}{\ln 3} - \frac{3}{\ln 3} \left[x^2 \frac{3^x}{\ln 3} - \int (x^2)' \frac{3^x}{\ln 3} dx \right]$$

Sol':

$f(x)$ and derivatives		$g(x)$ and antiderivatives
$\cdot x^3$ $f_0(x)$	$\xrightarrow{+}$	$\cdot 3^x$
$\cdot 3x^2$	$\xrightarrow{-}$	$\cdot \frac{1}{\ln 3} 3^x$ $g_1(x)$
$\cdot 6x$	$\xrightarrow{+}$	$\cdot \left(\frac{1}{\ln 3}\right)^2 3^x$ $g_2(x)$
$\cdot 6$	$\xrightarrow{-}$	$\cdot \left(\frac{1}{\ln 3}\right)^3 3^x$ $g_3(x)$
$\cdot 0$	$\xrightarrow{+}$	$\cdot \left(\frac{1}{\ln 3}\right)^4 3^x$ $g_4(x)$

Ans = $f_0(x)g_1(x) - f_1(x)g_2(x) + f_2(x)g_3(x) - \dots$

$$= x^3 \frac{1}{\ln 3} 3^x - 3x^2 \left(\frac{1}{\ln 3}\right)^2 3^x + 6x \left(\frac{1}{\ln 3}\right)^3 3^x - 6 \left(\frac{1}{\ln 3}\right)^4 3^x + \underline{0}$$

$$= x^3 \frac{3^x}{\ln 3} - 3x^2 \frac{3^x}{(\ln 3)^2} + 6x \frac{3^x}{(\ln 3)^3} - 6 \frac{3^x}{(\ln 3)^4}$$

Trigonometric integrals

We will see how various trigonometric identities may simplify integrals.

① $\int \sin^m x \cos^n x dx$, $(m, n \in \mathbb{N} \triangleq \{0, 1, 2, 3, \dots\})$.

(i) If m or n is odd, consider one of them, and take out

all the even powers. Then use $\sin^2 x + \cos^2 x = 1$.

e.g. $\int \sin^5 x \cos^7 x \, dx$

$$= \int \sin^4 x \cos^7 x (\sin x \, dx)$$

$$= - \int \sin^4 x \cos^7 x \, d\cos x$$

$$= - \int (1 - \cos^2 x)^2 \cos^7 x \, d\cos x$$

$$\underline{y = \cos x} \quad - \int (1 - y^2)^2 y^7 \, dy \mid y = \cos x$$

$$= - \int (y^7 - 2y^9 + y^{11}) \, dy \mid y = \cos x$$

$$= - \left(\frac{1}{8} y^8 - \frac{1}{5} y^{10} + \frac{1}{12} y^{12} \right) \mid y = \cos x + C.$$

□

e.g. $\int \cos^7 x \, dx$

$$= \int \cos^6 x \, d\sin x$$

$$= \int (1 - \sin^2 x)^3 \, d\sin x$$

$$\underline{y = \sin x} \quad \int (1 - y^2)^3 \, dy \mid y = \sin x$$

$$= \int (1 - 3y^2 + 3y^4 - y^6) \, dy \mid y = \sin x$$

$$= \int (1 - 3y^2 + 3y^4 - y^6) \, dy \mid y = \sin x$$

$$= \dots$$

□

(ii) If both m and n are even, then use the half-angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{and} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

e.g. $\int \sin^2 x \cos^4 x \, dx$

$$= \int \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$\frac{y=2x}{x=\frac{y}{2}} \int \frac{1 - \cos y}{2} \cdot \left(\frac{1 + \cos y}{2} \right)^2 d\left(\frac{y}{2}\right) \Big|_{y=2x}$$

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} \int (1 - \cos y) (1 + \cos y)^2 dy \Big|_{y=2x}$$

$$= \frac{1}{16} \int (1 + 2\cos y + \cos^2 y - \cos y - 2\cos^2 y - \cos^3 y) dy \Big|_{y=2x}$$

$$= \dots$$

$$= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx$$

$$\cdot \int 1 + \cos 2x \, dx = \dots = x + \frac{1}{2} \sin 2x + C_1.$$

$$\cdot \int \cos^2 2x \, dx = \int \frac{1 + \cos 4x}{2} dx = \dots = \frac{1}{2} x + \frac{1}{8} \sin 4x + C_2$$

$$\cdot \int \cos^3 2x \, dx \stackrel{u=2x}{=} \frac{1}{2} \int \cos^3 u \, du \Big|_{u=2x}$$

$$= \frac{1}{2} \int \cos^2 u \, d(\sin u) \Big|_{u=2x}$$

$$= \frac{1}{2} \int (1 - \sin^2 u) \, d(\sin u) \Big|_{u=2x}$$

$$= \dots$$

$$= \frac{1}{2} (\sin 2x - \frac{1}{3} \sin^3 2x) + C_3.$$

Combining and simplifying, we have

$$\text{LHS} = \frac{1}{8} \left(\frac{1}{2}x - \frac{1}{8}\sin 4x + \frac{1}{6}\sin^3 2x \right) + C. \quad [9]$$

$$(2) \int \tan^m x \, dx \quad (m \in \mathbb{N})$$

Sol: $m=0$ ✓

$$m=1. \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$$

$$= -\int \frac{1}{\cos x} \, d \cos x$$

$$= \ln |\sec x| + C.$$

$$m \geq 2. \text{ make use of } \begin{cases} \sec^2 x = \frac{\sin^2 x + \cos^2 x}{\cos^2 x} & = \tan^2 x + 1 \quad (1) \\ d \tan x = \frac{1}{\cos^2 x} \, dx & = \sec^2 x \, dx \quad (2) \end{cases}$$

$$\int \tan^m x \, dx = \int \tan^{m-2} x \cdot \tan^2 x \, dx$$

$$= \int \tan^{m-2} x (\sec^2 x - 1) \, dx$$

$$= \int \tan^{m-2} x \sec^2 x \, dx - \int \tan^{m-2} x \, dx$$

$$= \int \tan^{m-2} x \, d \tan x - \int \tan^{m-2} x \, dx$$

$$\stackrel{u=\tan x}{=} \int u^{m-2} \, du \Big|_{u=\tan x} - \int \tan^{m-2} x \, dx$$

$$= \frac{1}{m-1} u^{m-1} \Big|_{u=\tan x} - \int \tan^{m-2} x \, dx$$

$$= \frac{1}{m-1} (\tan x)^{m-1} - \int \tan^{m-2} x \, dx$$

• Step by step, you solve $\int \tan^m(x) \, dx$ from $\int \tan^{m-2}(x) \, dx$,
 from $\int \tan^{m-4}(x) \, dx$, ..., from $\int \tan^1(x) \, dx$ or $\int \tan^0(x) \, dx$.
 $= \int 1 \, dx$

[Numerical example : 8.3, example 5].

③ $\int \sec^n x \, dx \quad (n \in \mathbb{N})$.

— Assume $n \geq 3$.

$$\text{LHS} = \int \sec^{n-2} x \cdot \sec^2 x \, dx = \int \sec^{n-2} x (\tan x)' \, dx$$

$$= \sec^{n-2} x \cdot (\tan x) - \int (\sec^{n-2} x)' \tan x \, dx$$

$$= \sec^{n-2} x \cdot \tan x - \int (n-2) \sec^{n-3} x \cdot \frac{\tan x}{\cos x} \cdot \tan x \, dx$$

$$= \sec^{n-2} x \cdot \tan x - \int (n-2) \sec^{n-2} x \cdot \tan^2 x \, dx$$

$$\text{formula ①} \Rightarrow \sec^{n-2} x \cdot \tan x - \int (n-2) \sec^{n-2} x (\sec^2 x - 1) \, dx$$

$$= \sec^{n-2} x \cdot \tan x - \int (n-2) (\sec^n x - \sec^{n-2} x) \, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) \left[\underbrace{\int \sec^n x \, dx}_{\text{LHS}} - \int \sec^{n-2} x \, dx \right]$$

$$(\sec x)' = \left(\frac{1}{\cos x} \right)'$$

$$= +(\cos x)^{-2} (\sin x)$$

$$= \frac{\tan x}{\cos x}$$

✓

$\Rightarrow$

$$(n-1) \int \sec^n x \, dx = \sec^{n-2} x \cdot \tan x + (n-2) \cdot \int \sec^{n-2} x \, dx .$$

$\Rightarrow$

$$\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx .$$

Note that for even power n , using $\sec^2 x = 1 + \tan^2 x$ and making substitution $u = \tan x$ could be easier.

$$\begin{aligned} \int \sec^4 x \, dx &= \int \sec^2 x \sec^2 x \, dx \\ &= \int (1 + \tan^2 x) \, d \tan x \\ &= \int (1 + y^2) \, dy \big|_{y = \tan x} \\ &= \dots \end{aligned}$$